

# Nathan Cheng

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## EDUCATION

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**University of South Florida**, Tampa, FL

Aug 2021 – Dec 2025

B.S. Computer Science

– Data Structures & Algorithms, Operating Systems, Computer Architecture, Compilers, Linear Algebra, Analysis of Algorithms

## EXPERIENCE

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**Apple (Apple Vision Pro)**, Sunnyvale, CA

May 2024 – Aug 2024

Software Engineer Intern

- Engineered immersive AR/VR/XR video playback features for Apple Vision Pro on visionOS
- Developed with Swift, SwiftUI/UIKit, networking, and media streaming components in a large-scale, production codebase.
- Reduced media latency by 40% through asynchronous preprocessing, optimized streaming, and efficient metadata handling.
- Built XR interaction flows in collaboration with HCI and platform teams, ensuring low-latency, high-quality visual experiences
- Managed high-bandwidth camera and image/video data pipelines for realtime XR playback, processing, and rendering.

**Microsoft (Responsible AI)**, Redmond, WA

May 2025 – Aug 2025

Software Engineer Intern

- Developed an Azure-based Copilot agent that cut Responsible AI compliance work for Office Products by 25%.
- Extended RESTful APIs over HTTP with POST/GET endpoints for stateless agent-server communication.
- Applied prompt engineering to reduce hallucinations and improve response accuracy and consistency.

**MIT Lincoln Laboratory (OpenCV)**, Lexington, MA

Jan 2024 – Apr 2024

Spring Research Intern

- Deployed computer vision pipeline reducing modeling time by 80%, accelerating SysML workflows.
- Developed OpenCV-based pipeline using edge detection, Hough transform, and contour detection.
- Created CAD generation infrastructure for chip fabrication from provided design specs.
- Experience with C++ and on-device imaging workflows.

## PROJECTS

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**USublease — React, Google Maps API, AWS**

Oct 2025

- Launched campus-focused subleasing marketplace, reaching 10,000 visits in the first two months.
- Integrated AWS Lambda + S3 for serverless media storage and Google Maps Platform for geospatial UI.
- Optimized load performance with image compression and lazy-loading strategies.

**Ducky Beat Maker — Swift, SwiftUI**

Feb 2024

- Developed interactive mobile rhythm game for children featuring animated playable characters.
- Leveraged synchronous Swift + AVFoundation audio frameworks for real-time sound playback.
- Experience with iOS app development, SwiftUI/UIKit, and on-device audio processing.

**Fire Bending — Spectacles, Lens Studio, Snap AR**

Jul 2025

- Built immersive AR experience for Snap Spectacles enabling gesture-controlled fireball interactions.
- Implemented particle systems, physics-driven animations, and real-time rendering.
- Experience with XR pipelines, image data processing, and ML-enabled gesture recognition (SnapML).

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript, Java, Swift, C++, TypeScript, Flutter, SQL

**Frameworks:** ARCore, RealityKit, ARKit, AVKit, SwiftUI/UIKit, CoreBluetooth, TensorFlow, TensorFlow Lite, OpenCV, CoreML, Metal, MediaPipe, Snap AR SDK, SnapML, React, Video Encoding/Playback

**Platforms:** visionOS, iOS, Android, Spectacles, Azure, AWS (Lambda, S3), GCP, Lens Studio, Xcode

**Tools:** Git, Firebase, Jira, Postman, Pandas, SQL/NoSQL, Confluence, NDK, OpenGL

**Expertise:** iOS Development (2+ years), Swift, XR/AR/VR, Camera Systems, Image Data Processing, Realtime Rendering, On-device ML, Gesture Tracking, Scalable Backend, Full Stack